Take-home exam: December 18, 2014.

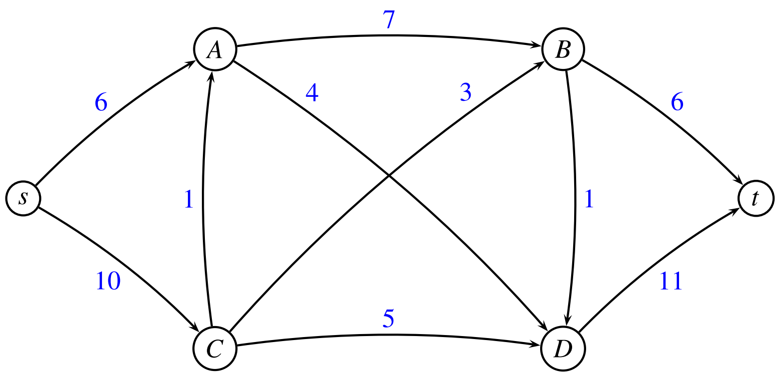
Course name: Network Optimization **Course no:** 42115

Rules: You are allowed to use your books, notes, slides, calculator, LP-solver etc. You are *not* allowed to get help from other people. You *must* sign the front page (found at the end of the assignment) and hand it in with your answer. You can write in Danish or English. It is allowed to write by hand and scan the solution. The solution should be uploaded at campusnet 4 hours after the project assignment has been posted.

Assignment: The assignment consists of 12 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found on the last pages of this assignment. You can print the figures, and use them in your answer by drawing on the figure.

Points: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 32 points can be obtained.

Maximum flow



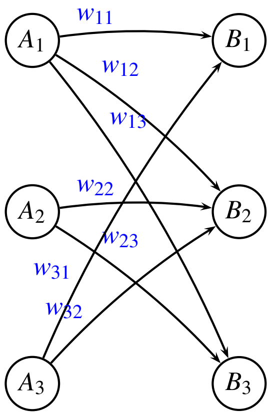
Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration use the augmenting path having *largest possible capacity*. Report which augmenting paths were used and their capacity. ■

Problem 2: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Bipartite graph

A bipartite graph is a graph whose vertices can be divided into two disjoint sets A and B such that every edge connects a vertex in A to a vertex in B.

A perfect matching is a matching which matches all vertices of the graph, i.e. every vertex of the graph is incident to exactly one edge of the matching.



Problem 3: (medium) Suppose we have a bipartite graph $G = (V,E)$ in which each edge e has a weight w_e. Weights are positive integers. Explain how we can solve the problem to find a perfect matching of maximum total value, if it exists, with help of minimum cost flow algorithms. (Hint: draw a small example.) ■

Problem 4: (medium) Suppose that we want to find a matching of maximum total weight, i.e., we allow that some vertices are not matched. Is it possible to find such a matching with just one run of a minimum cost flow algorithm? Explain why (not). ■

Modeling a hub port

In the project assignment [1] five different ways of modelling a hub port are presented. The five models can also be seen in the appendix at the end of this assignment. It is assumed that all vessels arrive at different days, and that they stay in the port during day time. In model 5 we assume that the vessels are sorted according to their time of arrival.

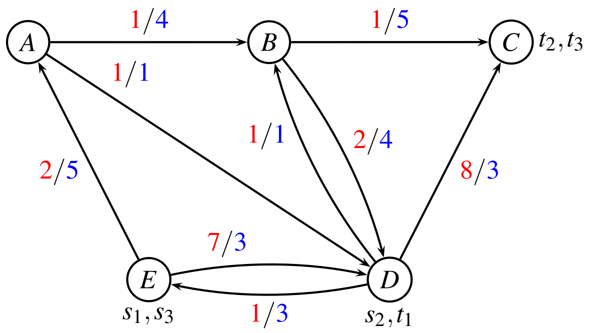
Problem 5: (small) Assume that at most C containers can stand on land from day to day during transshipments. The capacity C is shared among all commodities standing on land. Which of the five models can handle this constraint? Explain why (not). (It is not allowed to assign 0 capacity to some edges to "remove" them, since they may be carrying some other information). ■

Problem 6: (small) Assume that a tax has to be paid when containers are being transshipped from one vessel to another vessel. The tax depends on the nationality of the involved vessels. For instance a container being transshipped from a Danish vessel to a HongKong vessel need to pay a tax $T_{DK,HK}$. Which of the five models can handle such a tax? Explain why (not). ■

Unsplittable multicommodity flow problem

Consider the unsplittable multicommodity flow problem from the project assignment [1].

ASIANET (Route net of a liner shipping company)



k	s_k	t_k	d_k
1	E	D	2
2	D	C	2
3	E	C	1

For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. The right table describes the commodities k, with source s_k, terminal t_k and demand d_k.

We use *column generation* to solve the LP-relaxed path formulation of ASIANET, starting from the following routes:

commodity k	route	cost
1	E A B D	10
1	E D	14
2	D E A B C	10
3	E A B C	4

This leads to the following *restricted master problem*

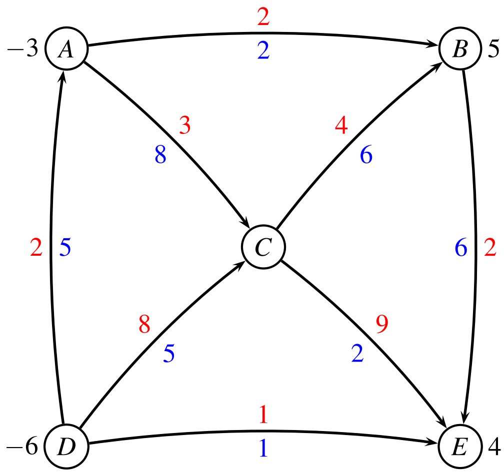
$$
\begin{array}{llllllll}
\min & 10x_1^1 & +\ 14x_2^1 & +\ 10x_1^2 & +\ 4x_1^3 & & & \\
\text{s.t.} & 2x_1^1 & & +\ 2x_1^2 & +\ 1x_1^3 & \leq 4 & & (y_{AB}) \\
 & & & & & \leq 1 & & (y_{AD}) \\
 & & & 2x_1^2 & +\ 1x_1^3 & \leq 5 & & (y_{BC}) \\
 & 2x_1^1 & & & & \leq 4 & & (y_{BD}) \\
 & & & & & \leq 1 & & (y_{DB}) \\
 & & & & & \leq 3 & & (y_{DC}) \\
 & & & 2x_1^2 & & \leq 3 & & (y_{DE}) \\
 & 2x_1^1 & & +\ 2x_1^2 & & \leq 5 & & (y_{EA}) \\
 & & 2x_2^1 & & & \leq 3 & & (y_{ED}) \\
 & 1x_1^1 & +\ 1x_2^1 & & & \geq 1 & & (\overline{y}_1) \\
 & & & x_1^2 & & \geq 1 & & (\overline{y}_2) \\
 & & & & x_1^3 & \geq 1 & & (\overline{y}_3) \\
\multicolumn{8}{l}{0 \leq x_1^1, x_2^1, x_1^2, x_1^3 \leq 1}
\end{array}
\qquad (1)
$$

Solving the above problem we get the primal solution $x_1^1 = \frac{1}{2}, x_2^1 = \frac{1}{2}, x_1^2 = 1, x_1^3 = 1$. The dual variables are $y_{AB} = -2$, and $\overline{y}_1 = 14$, $\overline{y}_2 = 14$, $\overline{y}_3 = 6$. All other dual variables are 0.

Problem 7: (medium) What is the next column that should be added to the model? (give sufficient details to justify your choice) ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Having solved a maximum flow problem to find a feasible solution we are told that $L = \{(D,C)\}$ and $U = \{(D,E),(A,B),(C,E)\}$ and $T = \{(D,A),(A,C),(C,B),(B,E)\}$. The flow on edges in U is at the upper bound, the flow in edges L is at the lower bound, while the edges in T form a tree.

Problem 8: (small) Calculate the flow corresponding to the above feasible solution defined by T,U,L. ■

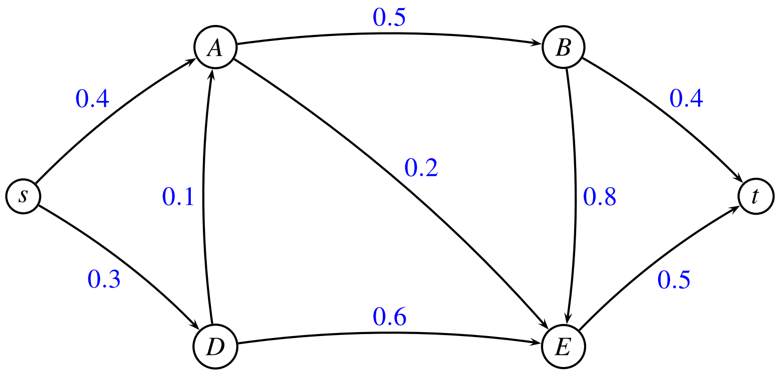
Problem 9: (large) Use the network simplex algorithm to solve the min cost flow problem to optimality. ■

Shortest path

In a directed communication network, a message is sent from a node s to another node t. At each arc (i,j) of the network there is a probability p_{ij} that the message will get lost on this arc. The probabilities p_{ij} are fixed and independent. An internet provider wants to choose a path from s to t, such that the probability that the message gets lost on this path is minimal. The probability that a message is *not* lost along a path using two edges (i,j) and (j,k) is $(1-p_{ij})\cdot(1-p_{jk})$.

Problem 10: (medium) Argue that a path from s to t minimizing the probability of being lost can be found by solving a shortest path problem from s to t where the edge weights w_{ij} are chosen as $w_{ij} = -\log(1-p_{ij})$ for edge (i,j). Hint: $\log(x\cdot y) = \log x + \log y$. ■

Problem 11: (small) Use the transformation for the following example where probabilities p_{ij} are marked with blue on the edges. It is sufficient to use two decimal digits precision in the transformation.



■

Problem 12: (medium) Choose an appropriate shortest path algorithm (argue why you choose it), and find the shortest path from s to t in the resulting graph. ■

end of assignment

References

[1] David Pisinger, "Flowing containers in liner shipping", *42115 Network Optimization, Project Assignment, DTU 2014*. can be downloaded from DTU CampusNet.

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Front page

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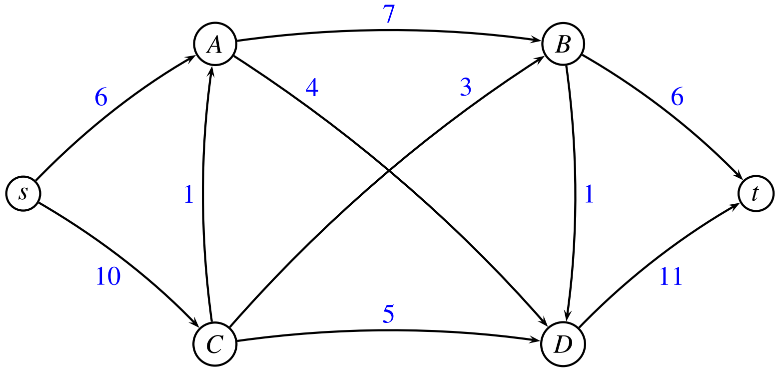
Name (printed letters)	study number

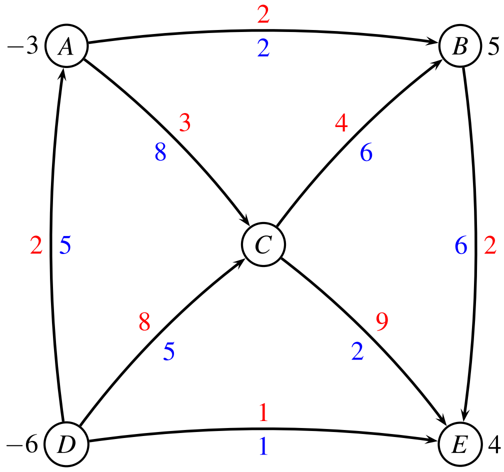
I hereby declare that I have worked alone on the exam and not discussed it with anybody else.

Date Signature Number of pages

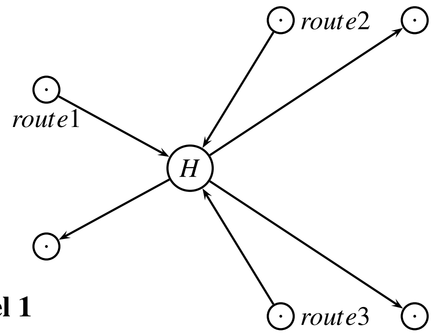
Maximum flow



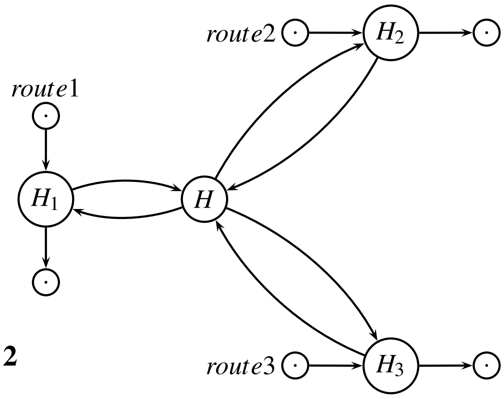
Min cost flow



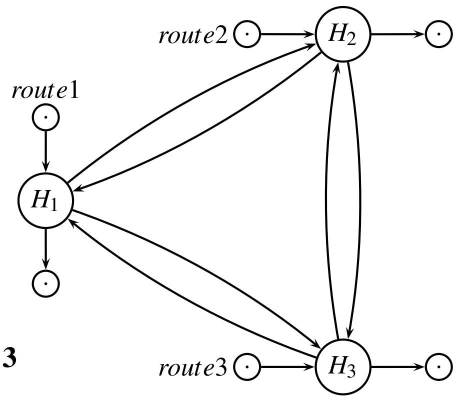
Modeling a hub port



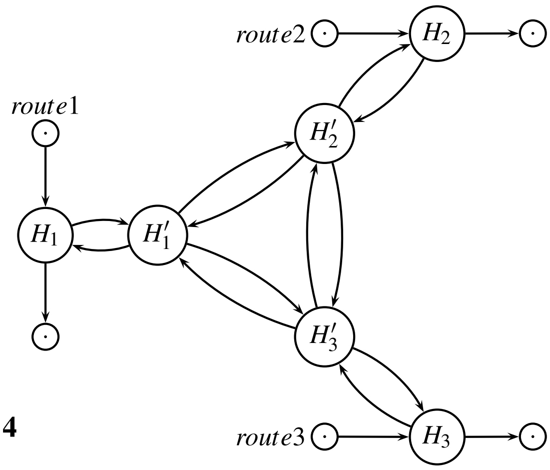
Model 1



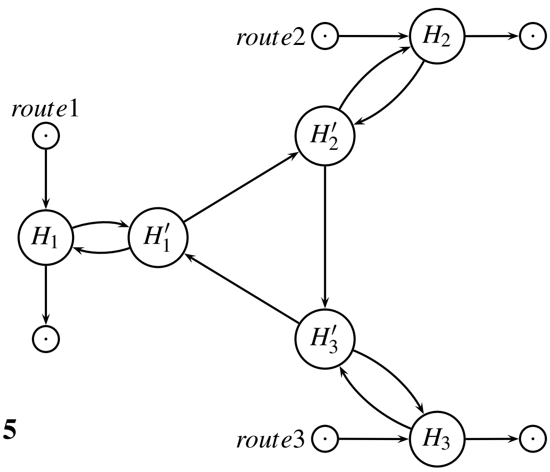
Model 2



Model 3

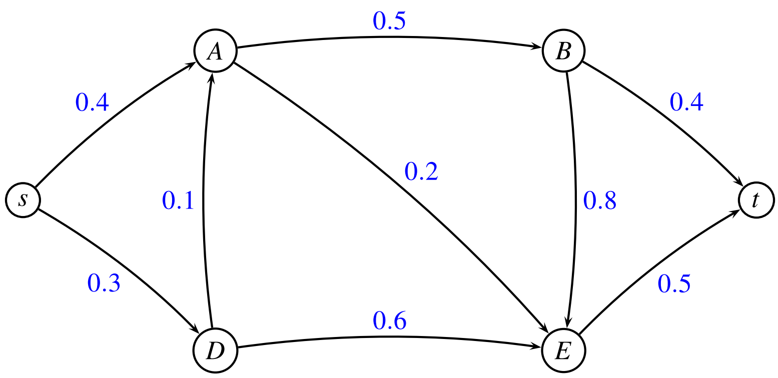


Model 4



Model 5

Shortest path



x	$\log x$
0.1	−1.00
0.2	−0.70
0.3	−0.52
0.4	−0.40
0.5	−0.30
0.6	−0.22
0.7	−0.15
0.8	−0.10
0.9	−0.05
1.0	0.00